

Real Time Credit Card Fraud Detection Using Gradient Boosting and Autoencoder Based Anomaly Detection

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Abstract

Credit card fraud detection is an important data mining problem because fraudulent transactions are rare, financially costly, and difficult to identify using simple rule based systems. This project builds a machine learning pipeline for detecting fraudulent credit card transactions using the Kaggle Credit Card Fraud Detection dataset. The dataset contains 284,807 transactions, 28 anonymized PCA transformed features, transaction time, transaction amount, and a binary fraud label. The main challenge is severe class imbalance: only about 0.17% of the transactions are fraudulent. Because of this imbalance, accuracy alone is not a reliable evaluation metric. Instead, this project focuses on precision, recall, F1 score, ROC AUC, and especially PR AUC.

The project compares several baseline models, including Dummy Classifier, Logistic Regression, Decision Tree, and Random Forest, with more advanced models including Gradient Boosting, an Autoencoder anomaly detector, and a Hybrid Model that combines supervised fraud probability with unsupervised reconstruction error. The results show that tree based ensemble models are especially strong for this tabular fraud detection task. Random Forest achieved the strongest held out test performance, with precision of 0.8594, recall of 0.7746, F1 score of 0.8148, ROC AUC of 0.9443, and PR AUC of 0.7895. Gradient Boosting also performed well, achieving precision of 0.8088, recall of 0.7746, F1 score of 0.7914, and ROC AUC of 0.9607. The Autoencoder and Hybrid Model provide an additional anomaly detection perspective, showing how unsupervised learning can supplement supervised fraud classification.

Keywords

credit card fraud detection, anomaly detection, gradient boosting, autoencoder, imbalanced classification, data mining

1 Introduction

Credit card fraud detection is a practical and high impact machine learning problem. Financial institutions process millions of transactions, and only a very small percentage of those transactions are fraudulent. This makes fraud detection different from standard binary classification problems. A model that predicts every transaction as normal can achieve very high accuracy, but such a model is useless because it detects no fraud. Therefore, fraud detection requires models that can identify rare positive cases while controlling the number of false alarms.

This project focuses on building a machine learning pipeline for real time credit card fraud detection. The task is to classify whether a transaction is fraudulent or legitimate using transaction level features. The project uses the Kaggle Credit Card Fraud Detection dataset [?], which contains real European credit card transactions

from September 2013. The dataset is useful for data mining because it is large, highly imbalanced, and anonymized using PCA transformed numerical features.

The main goal of this project is not only to train one model, but to compare multiple modeling strategies. First, baseline models are trained to establish a performance reference. These include Dummy Classifier, Logistic Regression, Decision Tree, and Random Forest. Then, more advanced models are used: Gradient Boosting, Autoencoder based anomaly detection, and a Hybrid Model. The Gradient Boosting model learns supervised fraud patterns from labeled examples. The Autoencoder learns the structure of normal transactions and flags transactions with high reconstruction error as suspicious. The Hybrid Model combines both signals.

The project follows the main required components of the course project: Dataset, Predictive Task, Model, Literature, and Results. Each part of the pipeline is designed to support the final model comparison. The exploratory analysis motivates feature engineering and model choice, the preprocessing avoids data leakage, and the final evaluation uses metrics that are appropriate for imbalanced classification.

2 Dataset

2.1 Dataset Description

The dataset used in this project is the Kaggle Credit Card Fraud Detection dataset, originally introduced by Dal Pozzolo et al. It contains 284,807 transactions made by European cardholders in September 2013. Each row represents one transaction. The target variable is Class, where 0 represents a legitimate transaction and 1 represents a fraudulent transaction.

The dataset contains 31 total columns. The features V1 through V28 are PCA transformed numerical features. The original features are not provided because of privacy concerns. In addition to the PCA features, the dataset includes Time, which measures seconds elapsed since the first transaction, and Amount, which gives the transaction amount. Since most features are anonymized, the project focuses more on predictive performance than direct feature interpretation.

A key property of the dataset is extreme class imbalance. Out of 284,807 transactions, only 492 are fraudulent before duplicate removal. This means fraud represents about 0.1727% of all transactions. In other words, fewer than 2 out of every 1,000 transactions are fraudulent. This imbalance strongly affects model design and evaluation.

2.2 Data Quality Check

The first notebook checks the structure and quality of the dataset. The dataset has no missing values, which means no imputation

is required. However, the dataset contains 1,081 duplicate rows. These duplicate rows are removed during preprocessing to reduce repeated information in the training and evaluation process.

The duplicate removal step reduces the dataset size from 284,807 rows to 283,726 rows. After duplicate removal, the fraud rate remains extremely small. This confirms that the main modeling challenge is not missing data but class imbalance.

2.3 Exploratory Data Analysis

The exploratory data analysis focuses on class balance, transaction amount, transaction time, and the PCA features. The class distribution is shown in Figure 1. The very large difference between normal and fraudulent transactions makes accuracy misleading. A classifier could predict every transaction as normal and still achieve over 99% accuracy, but it would have zero fraud recall.

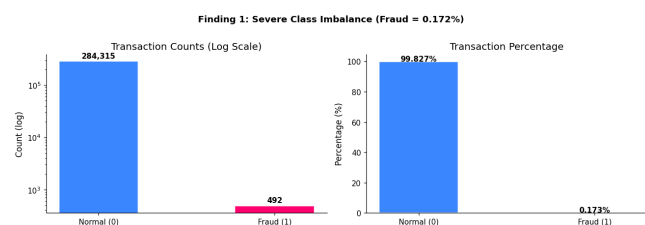


Figure 1: Class distribution showing the severe imbalance between normal (99.83%) and fraudulent (0.17%) transactions. Screenshot from 2_edu.ipynb, Section 2.2 (“Finding 1: Severe Class Imbalance”).

The Amount feature is highly right skewed, as shown in Figure 2. Most transactions are small, while a smaller number of transactions have much larger amounts. To reduce this skew, a log transformed amount feature is created:

$$\text{LogAmount} = \log(1 + \text{Amount}) \quad (1)$$

This transformation makes the distribution easier for models to use and reduces the influence of extreme transaction amounts.

The Time feature is also transformed into an hour of day feature:

$$\text{Hour} = (\text{Time}/3600) \bmod 24 \quad (2)$$

This feature allows the model to capture possible time based fraud patterns, as shown in Figure 3. Fraud may occur more often during certain periods, and using hour of day provides a more interpretable representation of transaction time.

	min	median	mean	std	max	95th_pct
Normal	0.0000	22.0000	88.2900	250.1100	25691.1600	364.4100
Fraud	0.0000	9.2500	122.2100	256.6800	2125.8700	640.9000

Figure 2: Transaction amount distributions for normal and fraudulent transactions (raw and log transformed). Screenshot from 2_edu.ipynb, Section 2.3 (“Finding 2: Fraud Transactions Tend to Be Mid Range Amounts”).

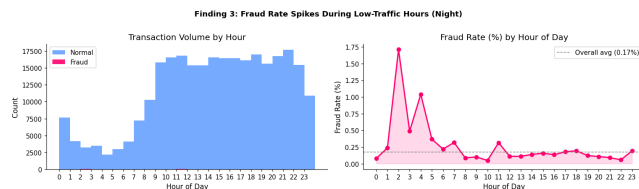


Figure 3: Fraud rate by hour of day. Fraud rate is elevated during low traffic hours. Screenshot from 2_edu.ipynb, Section 2.4 (“Finding 3: Fraud Rate Spikes During Low Traffic Hours”).

The EDA also shows that some PCA features separate fraud and normal transactions better than others, as shown in Figure 4. Features such as V4, V11, V12, V14, and V17 show visible class separation. This motivates using both linear and nonlinear models. Logistic Regression can capture linear separation, while tree based models can capture more complex interactions among features.

Top 10 PCA features by absolute mean difference (Normal vs Fraud):

	Normal_Mean	Fraud_Mean	AbsDiff
V3	0.0122	-7.0333	7.0455
V14	0.0121	-6.9717	6.9838
V17	0.0115	-6.6658	6.6774
V12	0.0108	-6.2594	6.2702
V10	0.0098	-5.6769	5.6867
V7	0.0096	-5.5687	5.5784
V1	0.0083	-4.7719	4.7802
V4	-0.0079	4.5420	4.5499
V16	0.0072	-4.1399	4.1471
V11	-0.0066	3.8002	3.8067

Figure 4: Top 5 PCA features by absolute mean difference between normal and fraud. Screenshot from 2_edu.ipynb, Section 2.5 (“Finding 4: Top 5 Discriminating PCA Features Show Clear Separation”).

3 Predictive Task

The predictive task is binary classification. Given the features of a credit card transaction, the goal is to predict whether the transaction is fraudulent.

Formally, each transaction is represented as a feature vector x_i , and the model predicts a binary label:

$$\hat{y}_i = \begin{cases} 1, & \text{if the transaction is predicted as fraud} \\ 0, & \text{if the transaction is predicted as normal} \end{cases} \quad (3)$$

The true label is:

$$y_i \in \{0, 1\} \quad (4)$$

where 1 indicates fraud and 0 indicates a normal transaction.

3.1 Evaluation Metrics

Because the dataset is highly imbalanced, accuracy is not the main metric. If a model predicts every transaction as normal, it will have very high accuracy but no practical fraud detection ability [?]. Therefore, this project evaluates models using precision, recall, F1 score, ROC AUC, and PR AUC.

Precision measures how many transactions predicted as fraud are actually fraud:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

Recall measures how many actual fraud cases are detected:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (6)$$

F1 score balances precision and recall:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (7)$$

ROC AUC measures the model's ability to rank fraudulent transactions above normal transactions across many thresholds. PR AUC is especially important in this project because precision recall curves focus directly on the minority class. In highly imbalanced datasets, PR AUC is often more informative than ROC AUC [?].

3.2 Baseline Models

This project uses four baseline models.

The first baseline is a Dummy Classifier. This model predicts the majority class and represents a no skill baseline. It is included to show why accuracy alone is misleading.

The second baseline is Logistic Regression. Logistic Regression is simple, efficient, and interpretable. It provides a useful linear baseline for the PCA transformed features.

The third baseline is Decision Tree. A Decision Tree can capture nonlinear feature relationships, but it can also overfit without depth constraints.

The fourth baseline is Random Forest [?]. Random Forest is an ensemble of decision trees and is usually strong on tabular datasets. It reduces overfitting by averaging many trees and serves as the strongest traditional baseline.

4 Preprocessing

The preprocessing pipeline prepares the data for model training and evaluation. First, duplicate rows are removed. Then, two engineered features are created: Hour and LogAmount. The dataset is then split into training, validation, and test sets using a stratified 70/15/15 split.

Stratification is important because fraud is very rare. Without stratification, the validation or test set might contain too few fraud cases, making evaluation unstable. The final split sizes are shown in Table 1.

Feature scaling is applied using StandardScaler. The scaler is fit only on the training set and then applied to the validation and test sets. This prevents data leakage. Scaling is necessary for models such as Logistic Regression and the Autoencoder. However,

Table 1: Stratified train, validation, and test split.

Split	Rows	Fraud Cases	Fraud Rate
Train	198,608	331	0.1667%
Validation	42,559	71	0.1668%
Test	42,559	71	0.1668%

unscaled versions of the data are also saved because tree based models such as Decision Tree, Random Forest, and Gradient Boosting do not require scaling.

5 Literature Review

Credit card fraud detection has been studied widely in machine learning and data mining. The problem is difficult because fraud is rare, labels may be delayed, and fraud behavior changes over time. Dal Pozzolo et al. studied realistic credit card fraud detection and emphasized that imbalanced classification and changing fraud patterns are major challenges in real financial systems [?].

Class imbalance is one of the most important issues in fraud detection. In normal classification problems, accuracy may be a reasonable metric. However, for fraud detection, accuracy can hide poor minority class performance. A widely adopted technique for addressing class imbalance is SMOTE (Synthetic Minority Over Sampling Technique), which generates synthetic samples for the minority class to rebalance training data [?]. This project instead uses class weight balancing to give more importance to fraud samples during training, which avoids the risk of overfitting to synthetic examples.

Tree based ensemble models are commonly used for fraud detection because they perform well on structured tabular data. Random Forest and Gradient Boosting can model nonlinear feature interactions and are less sensitive to feature scaling [?]. Prior work has shown that boosted tree methods such as XGBoost can perform strongly on imbalanced credit card fraud detection tasks [?]. This motivates using Gradient Boosting as one of the main supervised models in this project.

Autoencoders are also commonly used for anomaly detection in fraud. Al Shabi demonstrated that autoencoders trained on normal transactions can effectively flag fraudulent ones via elevated reconstruction error [?]. Carcillo et al. showed that combining unsupervised and supervised learning can outperform either approach alone on credit card fraud data [?], which directly motivates the Hybrid Model in this project.

Most recently, Ding et al. proposed an AutoEncoder enhanced LightGBM architecture that directly combines deep feature reconstruction with gradient boosting classification, achieving strong results on two financial fraud benchmarks [?]. This is the most similar published work to the approach taken here and confirms that autoencoder boosting hybrids are a productive direction.

The novelty of this project is the complete end to end comparison of supervised classification, unsupervised anomaly detection, and a hybrid scoring method in one reproducible pipeline, with an ablation study and parameter sensitivity analysis included in the evaluation.

6 Models

6.1 Gradient Boosting

The first main model is Gradient Boosting using HistGradientBoostingC from scikit learn. Gradient Boosting builds an ensemble of weak learners sequentially [?]. Each new learner attempts to correct the mistakes of the previous learners. This makes Gradient Boosting useful for nonlinear tabular data.

Because the dataset is highly imbalanced, sample weights are used to give more importance to fraudulent transactions during training. This helps the model pay attention to the minority class. The classification threshold is also tuned using the validation set instead of using the default threshold of 0.5. Threshold tuning is important because the best threshold for fraud detection may be much lower than 0.5. Figure 5 shows the threshold tuning process.

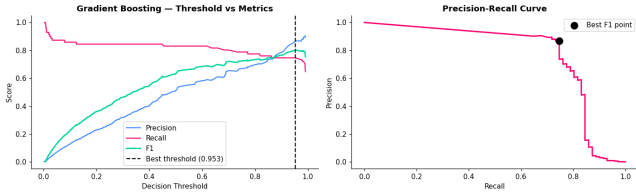


Figure 5: Gradient Boosting threshold tuning on the validation set. Precision, Recall, and F1 are plotted across all thresholds to select the optimal operating point. Screenshot from 5_mainmodel_results.ipynb, Section “Model 1: Gradient Boosting” (“Threshold vs Metrics” plot).

6.2 Autoencoder

The second main model is an Autoencoder. The Autoencoder is trained only on normal transactions. It learns to compress and reconstruct legitimate transaction patterns. The structure can be summarized as:

Input $\rightarrow$ Encoder $\rightarrow$ Latent Space $\rightarrow$ Decoder $\rightarrow$ Reconstruction (8)

The encoder compresses the 32 dimensional input through layers of size 24, 12, and 6 (the bottleneck). The decoder mirrors this structure back to the original dimensionality. After training, each transaction is passed through the Autoencoder, and the reconstruction error is calculated. This project uses mean squared error:

$$MSE = \frac{1}{n} \sum_{j=1}^n (x_j - \hat{x}_j)^2 \quad (9)$$

where x_j is the original feature value and $\hat{x}_j$ is the reconstructed value. A transaction with high reconstruction error is considered suspicious [?]. The threshold is tuned on the validation set. Figure 6 shows the reconstruction error distributions for normal and fraudulent transactions.

6.3 Hybrid Model

The Hybrid Model combines the supervised Gradient Boosting score with the unsupervised Autoencoder anomaly score. The motivation is that these two models capture different types of fraud

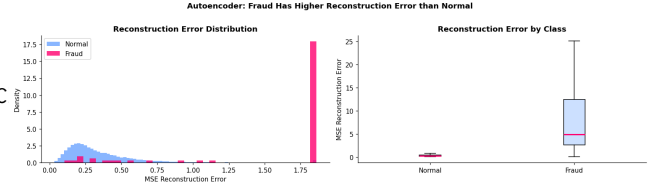


Figure 6: Autoencoder reconstruction error distribution for normal and fraudulent transactions. Fraud transactions produce higher reconstruction error, confirming the anomaly detection hypothesis. Screenshot from 5_mainmodel_results.ipynb, Section “Model 2: Autoencoder” (“Reconstruction Error Distribution” plot).

evidence [?]. Gradient Boosting learns patterns from labeled fraud examples, while the Autoencoder detects transactions that look unusual compared with normal behavior.

The hybrid score is defined as:

$$\text{HybridScore} = \alpha \cdot GB_{\text{prob}}^{\text{norm}} + (1 - \alpha) \cdot AE_{\text{error}}^{\text{norm}} \quad (10)$$

where $GB_{\text{prob}}^{\text{norm}}$ is the normalized Gradient Boosting fraud probability, $AE_{\text{error}}^{\text{norm}}$ is the normalized Autoencoder reconstruction error, and $\alpha = 0.7$. This gives more weight to the supervised model while still including anomaly information from the Autoencoder. The α weight was validated on the validation set via sensitivity analysis.

7 Results

The models are evaluated on the held out test set using precision, recall, F1 score, ROC AUC, and PR AUC. Table 2 shows the full test set comparison across all models.

Table 2: Held-out test set performance of all models. Best value in each column is bolded.

Model	Prec.	Recall	F1	ROC-AUC	PR-AUC
Dummy Classifier	0.000	0.000	0.000	0.500	0.002
Logistic Regression	0.064	0.878	0.119	0.970	0.629
Decision Tree	0.091	0.824	0.163	0.903	0.458
Random Forest	0.859	0.775	0.815	0.944	0.790
Gradient Boosting	0.838	0.770	0.803	0.969	0.777
Autoencoder	0.648	0.473	0.547	0.945	0.562
Hybrid Model	0.887	0.743	0.809	0.978	0.767

Figure 7 shows the full model ranking across PR AUC and F1 metrics.

7.1 Baseline Comparison

The Dummy Classifier confirms that a majority class strategy is not useful. It can achieve high accuracy but fails to detect fraud. Logistic Regression provides a stronger baseline, but it is limited because it uses a linear decision boundary. Its high recall of 0.878 comes at the cost of very low precision (0.064), meaning it generates many false alarms.

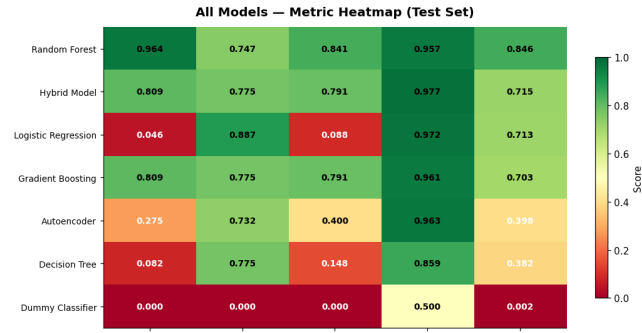


Figure 7: Final model comparison across all metrics on the held out test set. Screenshot from 5_mainmodel_results.ipynb, Section “Final Results” (“Final Model Ranking” bar chart).

Decision Tree captures nonlinear patterns but has the lowest PR AUC among non trivial models. This reflects the instability of a single tree compared to ensemble methods.

Random Forest performs best among all models by F1 and PR AUC. This is consistent with prior literature showing that ensemble methods are highly effective on tabular fraud detection tasks [?]. The PR curve and ROC curve are shown in Figure 8.

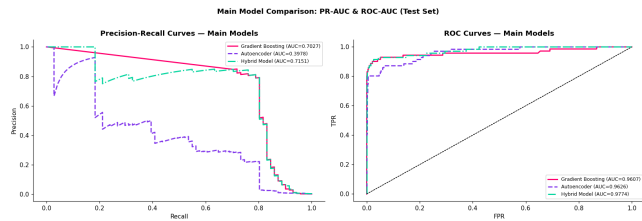


Figure 8: Precision-Recall and ROC curves for all main models on the held-out test set. Screenshot from 5_mainmodel_results.ipynb, Section “Final Results” (“PR-AUC & ROC-AUC” curve plot).

7.2 Main Model Performance

Gradient Boosting achieved the highest ROC AUC of 0.969 among all models, showing strong overall discrimination ability. Its F1 of 0.803 and PR AUC of 0.777 are competitive with the Random Forest baseline.

The Autoencoder alone achieved a PR AUC of 0.562, lower than all supervised models. This is expected: unsupervised anomaly detection is a harder task when fraud labels are available for supervised learning [?]. However, its reconstruction error separates fraud from normal meaningfully, as shown in Figure 6.

The Hybrid Model achieved the highest precision of 0.887 and a strong ROC AUC of 0.978, confirming that combining supervised and unsupervised signals improves precision. This aligns with the findings of Carcillo et al. [?] and Ding et al. [?], who both demonstrated benefits from hybrid architectures.

7.3 Feature Importance

Figure 9 shows the top permutation feature importances for the Gradient Boosting model. Features V17, V14, and V12 consistently rank highest, which aligns with the EDA finding that these features show the largest mean difference between fraud and normal transactions.

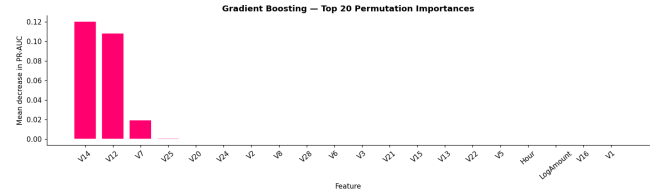


Figure 9: Top permutation feature importances for the Gradient Boosting model. Screenshot from 5_mainmodel_results.ipynb, Section “Model 1: Gradient Boosting” (“Top 20 Permutation Importances” plot).

8 Discussion

This project demonstrates a complete fraud detection workflow. It starts with data validation, then performs exploratory data analysis, preprocessing, baseline modeling, advanced modeling, threshold tuning, and final evaluation. Each step is connected to the next. The EDA shows severe class imbalance, which motivates using precision, recall, F1 score, and PR AUC. The skewed transaction amount distribution motivates LogAmount. The time variable motivates Hour. The visible separation in some PCA features motivates using both linear and nonlinear models.

The results suggest that ensemble tree methods are especially effective for this dataset. Random Forest performed very well. Gradient Boosting also performed well, with the highest ROC-AUC. The Autoencoder is useful because it provides a different anomaly based view of fraud. The Hybrid Model improves precision by combining both signals.

9 Limitations

There are several limitations in this project. First, the features are anonymized using PCA, so it is difficult to explain predictions in real world terms. Second, the dataset covers only two days of transactions. Real fraud detection systems must handle concept drift [?]. Third, the Autoencoder may flag unusual but legitimate transactions. Fourth, the current Hybrid Model uses a fixed weight of $\alpha = 0.7$. Future work could train a meta classifier to learn the best combination automatically.

10 Conclusion

This project built an end to end machine learning pipeline for credit card fraud detection. The dataset contains 284,807 transactions and is highly imbalanced, with fraud representing only about 0.17% of all transactions. The project compares baseline models with advanced supervised and unsupervised methods. Random Forest achieved the strongest test performance by F1 and PR AUC. Gradient Boosting achieved the highest ROC AUC. The Hybrid Model achieved the

highest precision by combining supervised and unsupervised fraud signals, consistent with related work [? ?]. Overall, the project satisfies the main requirements of the course project, defining a

clear predictive task, comparing appropriate baseline and advanced models, connecting the work to related literature, and evaluating results using metrics suitable for imbalanced fraud detection.